

## ONSLOW AP, Australia

WMO: 953050

Lat: 21.669S

Lon: 115.109E

Elev: 36

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 53.7       | 55.7 | 28.0                          | 22.0 | 77.2 | 31.3 | 25.6 | 75.6 | 23.8                  | 69.2 | 21.7 | 70.6 | 8.9                   | 150  | 0.543 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 16.6                   | 105.4           | 70.7 | 101.8 | 71.4 | 98.9 | 72.1 | 80.9                | 90.6 | 80.1 | 89.6 | 79.2 | 88.7 | 14.2                 | 210  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 78.5                            | 148.3 | 86.4 | 77.5 | 143.1 | 85.7 | 76.6 | 139.0 | 85.3 | 44.7          | 90.7 | 43.8 | 89.4 | 42.9 | 88.9 | 84.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 30.2          | 24.7 | 22.1 | 49.8                       | 114.9 | 1.7                | 2.3 | 48.6                                               | 116.6 | 47.7       | 117.9 | 46.7       | 119.2 | 45.5       | 120.9 |    |    |
| (5)               |      |      | 43.0                       | 82.7  | 1.5                | 1.0 | 41.9                                               | 83.5  | 41.0       | 84.1  | 40.2       | 84.6  | 39.1       | 85.4  |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 78.4  | 86.5  | 86.8  | 86.8  | 83.1  | 75.3  | 69.3  | 67.8  | 69.9  | 73.3  | 78.5  | 80.8  | 83.8  | (6)  |
| (7)  |                                                                                  | DBStd       | 8.03  | 4.82  | 4.28  | 3.93  | 3.89  | 3.93  | 3.63  | 3.22  | 3.30  | 3.45  | 4.94  | 4.67  | 5.06  | (7)  |
| (8)  |                                                                                  | HDD50       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  |
| (9)  |                                                                                  | HDD65       | 19    | 0     | 0     | 0     | 0     | 0     | 5     | 11    | 3     | 0     | 0     | 0     | 0     | (9)  |
| (10) |                                                                                  | CDD50       | 10381 | 1130  | 1030  | 1142  | 994   | 783   | 578   | 553   | 616   | 698   | 884   | 924   | 1049  | (10) |
| (11) |                                                                                  | CDD65       | 4925  | 665   | 610   | 677   | 544   | 318   | 133   | 98    | 154   | 249   | 419   | 474   | 584   | (11) |
| (12) |                                                                                  | CDH74       | 54996 | 8734  | 8113  | 8948  | 6094  | 2407  | 674   | 563   | 1062  | 2014  | 4265  | 5078  | 7044  | (12) |
| (13) |                                                                                  | CDH80       | 26979 | 4709  | 4381  | 4837  | 2975  | 837   | 94    | 78    | 229   | 687   | 2053  | 2448  | 3651  | (13) |
| (14) | Wind                                                                             | WSAvg       | 12.6  | 14.9  | 13.8  | 12.7  | 11.1  | 10.1  | 10.3  | 10.8  | 12.5  | 14.3  | 15.0  | 15.2  | (14)  |      |
| (15) | Precipitation                                                                    | PrecAvg     | 11.3  | 1.0   | 2.9   | 2.0   | 0.7   | 1.8   | 1.9   | 0.5   | 0.3   | 0.1   | 0.0   | 0.0   | (15)  |      |
| (16) |                                                                                  | PrecMax     | 39.3  | 5.7   | 21.5  | 14.5  | 4.6   | 9.0   | 8.9   | 3.5   | 2.4   | 1.0   | 1.1   | 0.9   | 0.7   | (16) |
| (17) |                                                                                  | PrecMin     | 3.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | (17) |
| (18) |                                                                                  | PrecStd     | 7.9   | 1.5   | 5.2   | 3.2   | 1.1   | 2.1   | 2.3   | 0.7   | 0.6   | 0.2   | 0.2   | 0.2   | 0.2   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 112.2 | 109.5 | 107.0 | 102.1 | 94.9  | 85.7  | 85.3  | 89.3  | 95.3  | 103.4 | 104.9 | 108.1 | (19) |
| (20) |                                                                                  |             | MCWB  | 69.9  | 70.9  | 71.6  | 68.6  | 67.1  | 64.6  | 61.5  | 61.2  | 64.1  | 65.6  | 67.2  | 69.3  | (20) |
| (21) |                                                                                  | 2%          | DB    | 104.1 | 102.6 | 102.1 | 98.4  | 90.2  | 82.7  | 82.1  | 85.0  | 90.2  | 98.0  | 99.5  | 102.3 | (21) |
| (22) |                                                                                  |             | MCWB  | 72.1  | 74.2  | 73.1  | 69.8  | 66.4  | 62.8  | 60.8  | 61.8  | 63.2  | 65.5  | 67.5  | 70.1  | (22) |
| (23) |                                                                                  | 5%          | DB    | 99.3  | 98.5  | 98.5  | 95.5  | 87.3  | 80.4  | 79.4  | 82.3  | 86.8  | 94.0  | 95.1  | 97.9  | (23) |
| (24) |                                                                                  |             | MCWB  | 74.5  | 75.4  | 73.9  | 70.1  | 65.8  | 61.8  | 60.5  | 61.1  | 62.9  | 66.2  | 68.4  | 71.6  | (24) |
| (25) |                                                                                  | 10%         | DB    | 95.4  | 95.1  | 95.4  | 92.7  | 84.8  | 78.1  | 77.2  | 79.9  | 83.8  | 90.5  | 91.3  | 94.2  | (25) |
| (26) |                                                                                  |             | MCWB  | 75.8  | 76.5  | 75.0  | 70.5  | 66.2  | 62.1  | 60.4  | 61.4  | 62.6  | 65.9  | 69.0  | 72.4  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 81.8  | 82.7  | 81.4  | 79.3  | 76.5  | 73.4  | 70.8  | 70.4  | 71.5  | 74.0  | 77.3  | 81.0  | (27) |
| (28) |                                                                                  |             | MCD B | 92.6  | 92.3  | 90.4  | 87.9  | 80.9  | 77.1  | 75.1  | 76.4  | 80.7  | 84.2  | 86.3  | 89.3  | (28) |
| (29) |                                                                                  | 2%          | WB    | 80.6  | 81.0  | 80.1  | 77.5  | 73.5  | 70.9  | 68.3  | 68.3  | 69.5  | 72.3  | 75.8  | 79.2  | (29) |
| (30) |                                                                                  |             | MCD B | 91.2  | 90.7  | 89.2  | 85.7  | 80.3  | 74.6  | 73.2  | 75.3  | 79.1  | 83.7  | 85.5  | 88.2  | (30) |
| (31) |                                                                                  | 5%          | WB    | 79.8  | 80.1  | 79.2  | 76.4  | 71.7  | 68.9  | 65.9  | 66.6  | 68.1  | 71.0  | 74.3  | 77.5  | (31) |
| (32) |                                                                                  |             | MCD B | 89.8  | 89.7  | 88.7  | 85.0  | 79.4  | 73.7  | 72.0  | 74.4  | 78.3  | 83.0  | 85.1  | 87.8  | (32) |
| (33) |                                                                                  | 10%         | WB    | 78.9  | 79.2  | 78.3  | 75.1  | 70.0  | 66.7  | 63.8  | 65.0  | 66.6  | 69.6  | 72.8  | 76.1  | (33) |
| (34) |                                                                                  |             | MCD B | 88.5  | 88.7  | 88.1  | 85.0  | 78.5  | 73.0  | 71.3  | 73.8  | 76.7  | 82.2  | 84.6  | 87.1  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 17.3  | 16.6  | 17.6  | 19.7  | 19.7  | 18.5  | 20.2  | 22.1  | 22.8  | 23.2  | 20.7  | 19.1  | (35) |
| (36) |                                                                                  |             | MCDBR | 25.1  | 22.5  | 22.3  | 23.1  | 22.7  | 20.9  | 23.3  | 25.2  | 26.8  | 28.1  | 26.5  | 25.8  | (36) |
| (37) |                                                                                  |             | MCWBR | 8.8   | 8.2   | 8.7   | 9.0   | 9.4   | 9.0   | 10.2  | 10.6  | 10.8  | 9.7   | 9.5   | 9.0   | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 16.1  | 15.9  | 16.7  | 18.0  | 17.9  | 13.5  | 16.7  | 20.9  | 22.0  | 21.9  | 19.1  | 17.0  | (38) |
| (39) |                                                                                  |             | MCWBR | 6.7   | 7.0   | 7.7   | 8.4   | 9.4   | 7.8   | 9.5   | 10.4  | 10.5  | 9.7   | 9.3   | 7.6   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.393 | 0.394 | 0.374 | 0.352 | 0.330 | 0.321 | 0.314 | 0.317 | 0.328 | 0.345 | 0.351 | 0.367 | (40)  |      |
| (41) |                                                                                  | taud        | 2.512 | 2.525 | 2.570 | 2.594 | 2.596 | 2.606 | 2.605 | 2.558 | 2.519 | 2.493 | 2.510 | 2.531 | (41)  |      |
| (42) |                                                                                  | Ebn at Noon | 302   | 298   | 296   | 290   | 284   | 281   | 287   | 297   | 305   | 309   | 313   | 310   | (42)  |      |
| (43) |                                                                                  | Edh at Noon | 36    | 35    | 32    | 29    | 27    | 26    | 26    | 30    | 33    | 36    | 36    | 36    | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2449  | 2317  | 2043  | 1722  | 1439  | 1289  | 1390  | 1735  | 2082  | 2376  | 2564  | 2594  | (44)  |      |
| (45) |                                                                                  | RadStd      | 118   | 129   | 128   | 97    | 104   | 90    | 104   | 57    | 68    | 65    | 70    | 108   | (45)  |      |

## Historical Trends

|                             | DBAvg | Heating |       |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |       | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)   | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           | N/A   | N/A     | N/A   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (6 neighbors) | +0.56 | N/A     | -1.84 | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +275  | +288  |

Nomenclature: See separate page